

Test Documentation: Scaling & Stress Audit

Overview

This suite evaluates the performance limits and reliability of the SDK under massive DAG loads, high I/O concurrency, and RPC bombardment.

Test Goals

1. **Iterative Traversal Verification:** Ensuring that extremely deep VT XO chains (Ancestor -> Child) do not cause `Stack Overflow`.
2. **Concurrency Control:** Throttling simultaneous on-chain requests to prevent node saturation.
3. **Verification Cache Performance:** Verifying that redundant validations for the same VT XO Root are cached and returned instantaneously.

Environment

- **Providers:** `MockIndexerProvider` and `MockOnchainProvider`.
- **Data Source:** Linear chains of virtual transactions with depths ranging from 10 to **10,000**.
- **Execution:** Node.js/Vitest environment with a 90-second timeout.

Evaluated Scenarios

Scenario	Description	Expected	Result
Deep Chain (10k)	A linear Reversed-DAG with 10,000 ancestor levels.	PASS	~90s (Iterative)
Concurrency (100x)	100 simultaneous verification calls for the same VT XO Root.	PASS	~3.3s (Cached)
I/O Flooding	500 simultaneous <code>onReceiveVtxo</code> calls.	PASS	Throttled

Results

- **Iterative Traversal:** Successfully prevented `RangeError: Maximum call stack size exceeded` in all 10k-node tests (87/87 tests total passed).
- **Cache Hit Rate:** Re-verification latency dropped from 90s (10k nodes) to <1ms on cache hit.
- **Resource Protection:** The `ConcurrencyLimiter` successfully queued and processed over 100 simultaneous RPC calls without worker starvation.